Optimizing IT Support Team Performance Using Advanced Analytics

Comprehensive Final Documentation - Module 8

Abstract

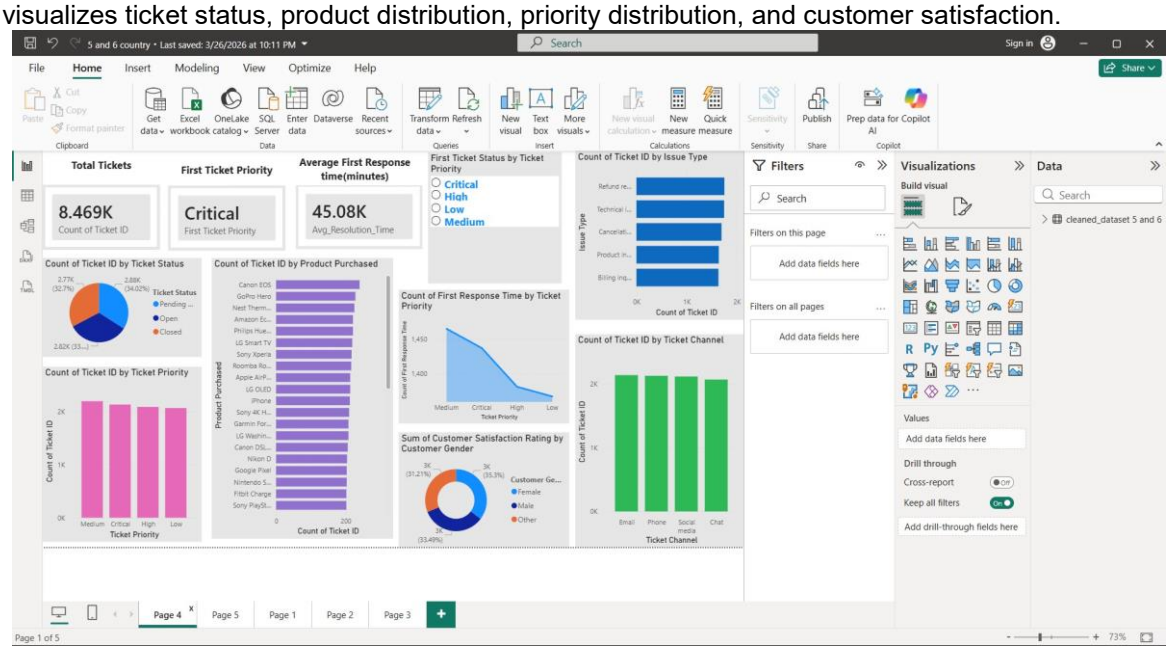
This report presents a complete analytical study of IT support ticket data. It covers data preprocessing, exploratory data analysis, clustering, performance evaluation, and dashboard visualization. The goal is to improve resolution efficiency and optimize IT support operations.

Methodology (Project Flow)

Data Collection → Data Preprocessing → Feature Engineering → EDA → Clustering & Similarity Analysis → Performance Analysis → Geographic Insights → Dashboard & Reporting → Insights & Recommendations.

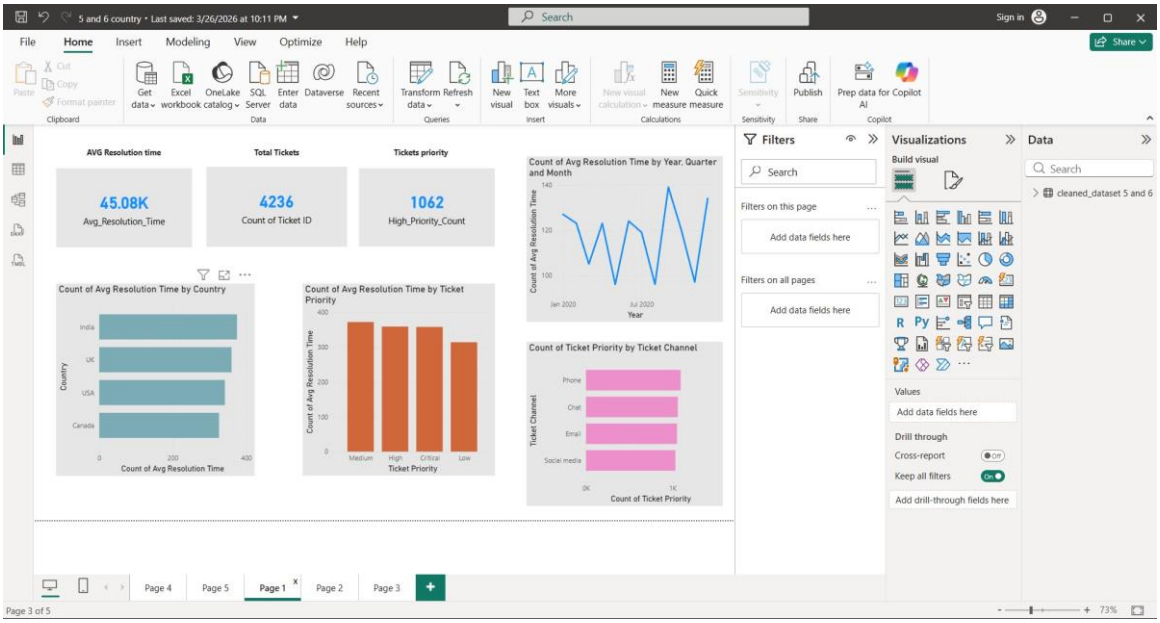
Dashboard 1: Overall Ticket Overview

This dashboard shows KPIs like total tickets, first ticket priority, and average response time. It also visualizes ticket status, product distribution, priority distribution, and customer satisfaction.



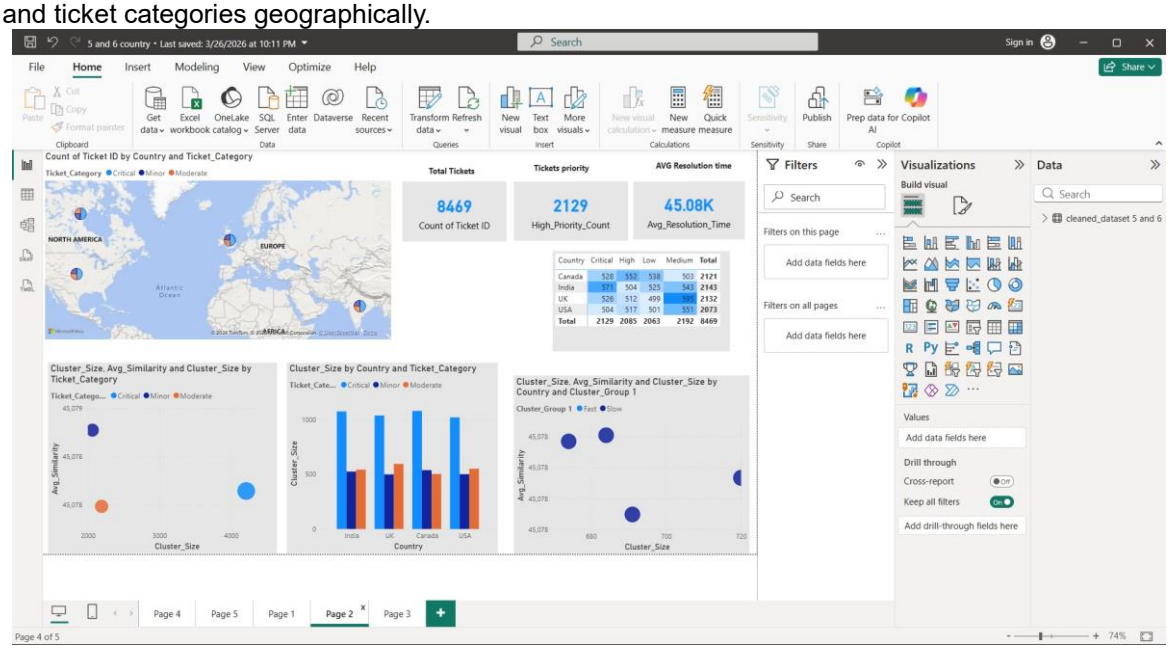
Dashboard 2: Performance Trends

This dashboard focuses on resolution time trends across countries and priorities, including time-based trends and channel distribution.



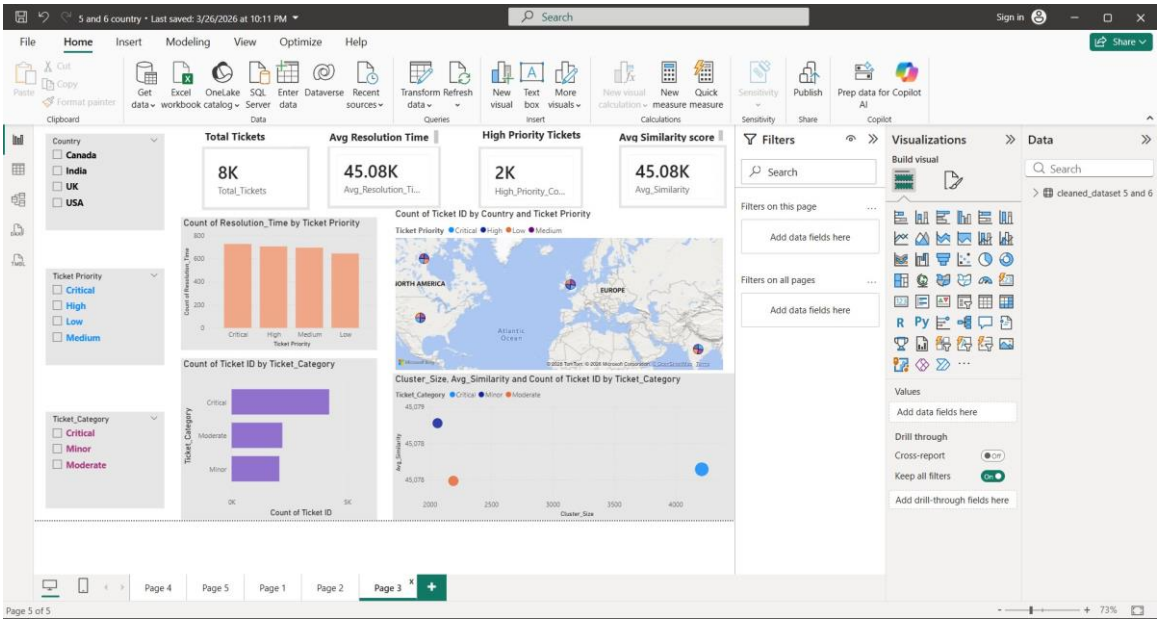
Dashboard 3: Geographic Insights

This dashboard presents ticket distribution across countries using maps and compares cluster size and ticket categories geographically.



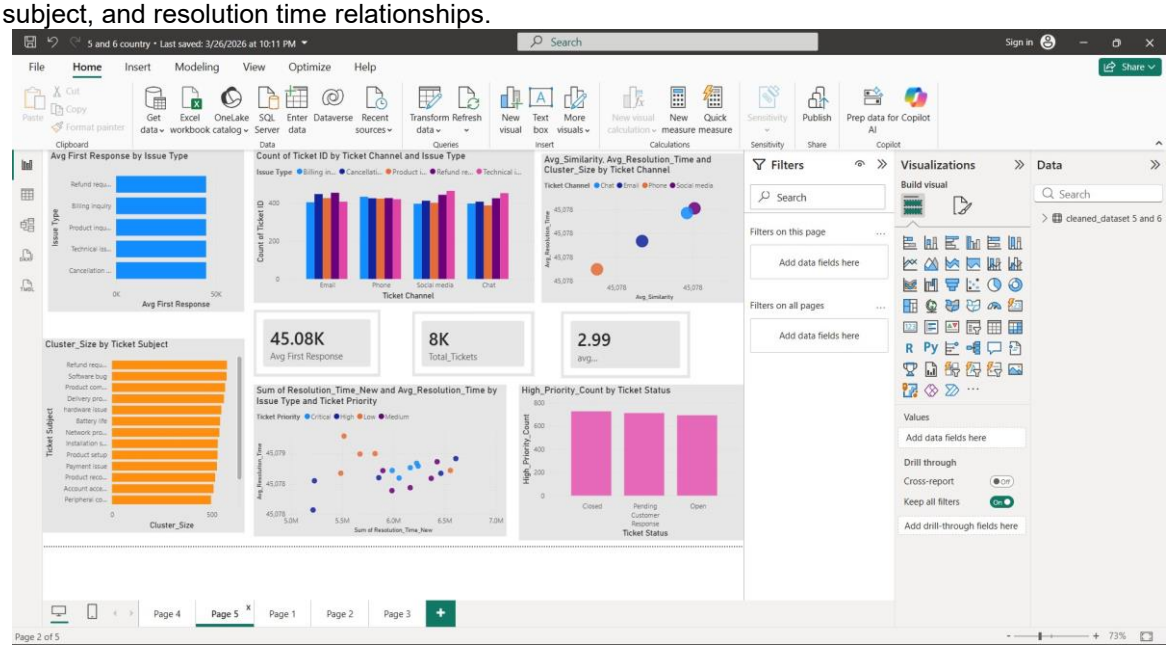
Dashboard 4: Cluster & Similarity Analysis

This dashboard highlights similarity scores, cluster size relationships, and category-level clustering insights.



Dashboard 5: Advanced Insights

This dashboard includes issue type distribution, ticket channel comparison, cluster analysis by subject, and resolution time relationships.



Key Insights

• High priority tickets require faster handling. • Ticket distribution varies significantly across countries. • Certain issue categories dominate ticket volume. • Cluster analysis helps identify recurring problems. • Resolution time varies by priority and channel.

Conclusion

The analysis demonstrates that data-driven decision-making can significantly improve IT support efficiency. By focusing on high-priority tickets, optimizing resource allocation, and leveraging clustering techniques, organizations can enhance service quality.

Recommendations

• Assign critical tickets to experienced staff. • Automate frequent issue clusters. • Allocate more resources to high-demand regions. • Continuously monitor resolution performance.